

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT INITIAL PROPOSAL

[Edu Path (Academic & Career Advisor)]

Project Team

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Project Proposal

Project Title: AI powered Academic and Career Decision Support System

Introduction:

The Process of selecting the appropriate University, Program, and Career is not an easy thing considering many factors involved like marks, interests, finances, and eligibility. The AI Powered Personalized Academic and Career Decision Support System utilizes AI and Machine Learning techniques to study all these factors. It suggests appropriate universities and programs according to the individual's profile. Our system provides eligibility checking, career pathways, skill gap analysis, and personalized explanations.

Problem Statement:

Choosing the right university, the course of study, and the career of choice becomes a challenge for students due to the lack of centralized information and several factors involved in making these choices like grades, interests, eligibility, budget, and career objectives. Current recommendation systems offer limited data about universities and courses without providing any personalized recommendations. There is a need for a smarter solution that could analyze the profile of the student and make appropriate recommendations.

Objectives:

1. To design a questionnaire for gathering students' academic background, interests, skills, career preferences, and other relevant data.
2. To train the existing language model using a student's database and enable it to recommend the most suitable fields of study.
3. To recommend most suitable degrees within the fields of study suggested by the language model.
4. To assess the student's eligibility for the suggested degrees and universities by comparing his/her academic qualifications with the admission requirements.
5. To filter out universities in accordance with the student's preferences, including the province and city of his/her choice, as well as his/her preferred

- type of universities (public or private), using the university database.
6. To prioritize the shortlisted universities in accordance with the student's preferences, including:
 - Budget.
 - Scholarships.
 - Distance and transportation facilities.
 7. To provide students with the most relevant and personalized degree recommendation, along with the rationale behind the recommendation. Similarly, recommend the most suitable universities within the selected specialization.

Proposed Solution:

- The proposed system is an AI-based personalized recommendation system that recommends academic fields and universities to the students.
- The students will provide their academic background, interests, skills, career interests, and university preferences in a questionnaire.
- A language model already trained on the relevant student database will be used to recommend academic fields of study to the students.
- The system will recommend the best degrees to pursue based on the recommended field of study, academic background, interests, skills, career interests, and eligibility of the student.
- The universities will be filtered using the Kaggle university database based on the location, the sector of the university, programs offered, and the eligibility of the student. The shortlisted universities based on the location, the sector of the university, programs offered, and the eligibility of the student will be ranked using the budget, scholarships, distance, and transport to create a recommendation list using TOPSIS.
- The final recommendation will provide the best field of study, degree program, and the best universities for the student.

Methodology

- **Data Collection:** Obtain a student dataset and a Kaggle dataset containing information about universities and their attributes.
- **Data Preprocessing:** Clean, transform, and standardize the datasets and prepare the student features required for training.
- **Model Training:** Train/fine-tune an existing language model using the student dataset to recommend suitable academic fields.
- **Program Recommendation:** Use the predicted academic field to identify

- suitable degree programs and check the student's eligibility.
- **University Filtering:** Filter universities based on the student's selected province, city, sector, program availability, and eligibility.
 - **University Ranking:** Apply **TOPSIS (MCDM)** to rank eligible universities using budget, scholarship availability, distance, and transportation facilities.
 - **Evaluation:** Evaluate the trained model and overall recommendation system using appropriate performance metrics and recommendation validation.

System Architecture and Workflow:

- The proposed solution includes a series of steps, namely:
- **Student Information Collection.** A student fills out the questionnaire, providing personal, academic, interest, skill, career, and university-related information.
- **Student Profile Generation.** The student's responses are gathered and organized into a structured student profile.
- **Academic Field Recommendation.** The student profile in a standard format is submitted to an existing language model fine-tuned on the student data to recommend an appropriate academic field of study.
- **Program Recommendation.** From the recommended academic field of study, the system recommends degree programs that fit the student's academic profile and checks their compatibility.
- **University Filtering.** The university data set from Kaggle is filtered based on the student's preference of province, city, university sector, available programs, and eligibility criteria using a constraint-based filtering approach.
- **University Ranking.** The shortlisted universities are ranked using the TOPSIS (MCDM) algorithm considering budget, scholarships, distance, and transportation.
- **Final Recommendation.** The system finally recommends to the student the most appropriate academic field of study, compatible degree programs, and the best-fit universities with relevant details.

Applications

- **Student Academic Guidance:** Help students identify suitable academic fields and degree programs based on their academic background, interests, and skills.
- **University Selection:** Assist students in selecting suitable universities based on program availability, eligibility, budget, scholarships, distance, and transportation facilities available.
- **Career and Education Counseling:** Provide academic counselors with information that can be used to advise students.

- The admission decision support system helps students find universities and programs for which they are qualified to apply. It then guides them through the process of choosing a degree program and university.
- This solution provides personalized guidance, from recommending an academic field of study through suggesting a degree program and university.
- Universities, colleges, and career counseling centers can use the system as a decision-support tool when advising prospective students.
- In the future, the system can also be expanded to include information on career options, postgraduate programs, overseas universities, and scholarships.

Advantages of the Proposed System:

- **Personalized Recommendations:**
The proposed system will recommend suitable academic fields, programs, and universities according to the student's preferences and eligibility.
- **Eligibility-Based Selection:**
The system will select programs and universities according to the eligibility of the student.
- **Multi-Criteria University Ranking:**
The system will rank the universities according to their eligibility, scholarships, distance, transportation, and budget (TOPSIS).
- **Complete Guidance From Academic Fields to Universities:**
The system will guide the students from choosing academic fields to recommending universities.
- **Better Decision Making:**
The system will help students take better decisions by comparing and contrasting the options available to them.

Limitations:

- **Dataset Coverage:** The system can only recommend the universities and programs that are covered by the provided dataset and their available web sources.
- **Web Data Availability:** The data depends upon the availability of the university websites for crawling.
- **Changing Information:** Even though the system updates university information automatically on a weekly basis, there could be some changes that the system might not detect before the next weekly update.
- **Model Limitations:** The accuracy of field of study recommendations depends on the quality of the training dataset and the performance of the trained language

model.

- **Limited Scope:** The initial system is limited to recommending undergraduate fields of study, programs, and universities in Pakistan.

Future Scope

- **Career Recommendation:** The system can be incorporated to recommend career guidance to students according to their academic data and inclinations.
- **Scholarship Recommendation:** The future scope of the system includes a scholarship recommendation module which will provide relevant scholarship opportunities to the students on the basis of their qualification and preferences.
- **International Universities:** The system can be modified to provide information about the universities abroad.
- **Postgraduate Programs:** The system can be upgraded with recommendations for postgraduate courses.
- **Real-Time Information:** The web-crawling mechanism of the system can be improvised and updated to provide the latest information regarding the universities.

Conclusion

The suggested system will offer tailored recommendations of academic areas of study, degrees, and universities based on personal student profiles, taking into account their preferences. To do this, an academic field recommendation system that has been trained on a student body database by using a language model will be applied. Additionally, eligibility-based filtering and TOPSIS algorithm will be applied to choose the appropriate programs and universities. Moreover, the system will use HTTP requests and web crawling to keep the university information updated. In general, the mentioned system will serve as a powerful instrument for personalized higher-educational decision-making.